Yiqi Liu

Contact Information

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Interests

Econometrics : causal inference, partial identification, machine learning/nonparametric methods

Education

Cornell University, Ithaca, NY

Ph.D. in Economics — Expected 2026

Tulane University, New Orleans, LA

B.S. in Mathematics and Economics with minor in Studio Art, *summa cum laude* — 2020

Working Papers

Synthetic Parallel Trends (Job Market Paper) [draft]

Abstract : Popular empirical strategies for policy evaluation in the panel data literature—including difference-in-differences (DID), synthetic control (SC) methods, and their variants—rely on key identifying assumptions that can be expressed through a specific choice of weights ω relating pre-treatment trends to the counterfactual outcome. While each choice of ω may be defensible in empirical contexts that motivate a particular method, it relies on fundamentally untestable and often fragile assumptions. I develop an identification framework that allows for all weights satisfying a *Synthetic Parallel Trends* assumption : the treated unit's trend is parallel to a weighted combination of control units' trends for a general class of weights. The framework nests these existing methods as special cases and is by construction robust to violations of their respective assumptions. I construct a valid confidence set for the identified set of the treatment effect, which admits a linear programming representation with estimated coefficients and nuisance parameters that are profiled out. In simulations where the assumptions underlying DID or SC-based methods are violated, the proposed confidence set remains robust and attains nominal coverage, while existing methods suffer severe undercoverage.

Inference for an Algorithmic Fairness-Accuracy Frontier (with Francesca Molinari) [draft]
- Revise and resubmit at the *American Economic Review*
- Extended abstract in *Proceedings of the 25th ACM Conference on Economics and Computation*

Using Forests in Multivariate Regression Discontinuity Designs (with Alice Y. Qi) [draft]

Work in progress

Partial Identification of Algorithmic Frontiers with Selective Labels (with Francesca Molinari and Amilcar Velez)

Professional Activities

Conference/Workshop Presentations
- 2025 : Econometric Society World Congress, Stanford Data-Driven Seminar
- 2024 : The 25th ACM Conference on Economics and Computation, Bravo/SNSF Workshop on Using Data to Make Decisions at Brown University, ESIF Economics and AI+ML Meeting

Journal Referee
Journal of Political Economy, Review of Economic Studies

Conference Reviewer
2025 ACM Conference on Fairness, Accountability, and Transparency (FAccT)

Other Service
Student organizer, Cornell Econometrics Reading Group — 2022-2024

Fellowships & Awards

Ernest Liu ′64, Ta-Chung and Ya-Chao Liu Memorial Fellowship, Cornell — Spring 2024
Tapan Mitra Memorial Prize for Outstanding Third Year Paper, Cornell — 2023
The Louis Walinsky Fund in Economics Outstanding Teaching Award, Cornell — 2022

Sage Fellowship, Cornell	2020-2025
Research Award in Economics, Tulane	2020
Phi Beta Kappa, Tulane	2020
Presidential Scholarship Award, Tulane	2016-2020

Research & Teaching Experience

Research Assistant for Levon Barseghyan and Francesca Molinari — 2022-2023, 2025-2026

Research Assistant for Francesca Molinari — Summer 2022, 2023, 2024

Teaching Assistant for Douglas McKee — Spring 2022

- ECON 3120 : Applied Econometrics, undergraduate (grading and office hours only)

Teaching Assistant for Suraj Malladi — Fall 2021

- ECON 6170 : Mathematical Economics, first-year Ph.D. core (evaluation : 4.93/5)

Advising Committee

Prof. Francesca Molinari (Chair)
Department of Economics
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Prof. Levon Barseghyan
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Prof. José Luis Montiel Olea
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